

Problem Set 3: Solving Linear Equations

Mathematics 8 - Narra | Total: 30 points

Directions: Copy each item, then show your complete solution. Encircle the final answer.

Submission: handwritten, before our next meeting. Answers without solutions earn no credit.

Part A. Two-Step Equations (2 points each)

- 1. $x + 7 = 15$
- 2. $4x = 36$
- 3. $2x - 5 = 11$
- 4. $x/3 + 4 = 9$
- 5. $5x + 12 = 47$
- 6. $18 - 3x = 6$

Part B. Variables on Both Sides (3 points each)

- 7. $5x = 2x + 21$
- 8. $7x - 4 = 3x + 12$
- 9. $9 - 2x = 4x - 15$
- 10. $6(x - 2) = 3x + 9$

Part C. Word Problems (5 points each)

- 11. Marco is three times as old as his sister. The sum of their ages is 24. How old is each of them?
- 12. A rectangle is twice as long as it is wide. Its perimeter is 36 meters. Find its dimensions.

Answer Key

Teacher copy only - detach this page before printing for the class.

- 1. $x = 8$
- 2. $x = 9$
- 3. $x = 8$
- 4. $x = 15$
- 5. $x = 7$
- 6. $x = 4$
- 7. $x = 7$
- 8. $x = 4$
- 9. $x = 4$
- 10. $x = 7$
- 11. Sister is 6 years old; Marco is 18 years old.
- 12. Width = 6 m; Length = 12 m.